

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
Student Name:	B.VANI	Reg. No.:	192472348
Section / Batch:	CSE-AI/2024	Date:	

Code of Conduct

I B.VANI(192472348) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: B.VANI

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

1. Reference Resolution and Discourse Coherence

Given the following paragraph:

"Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving."

Tasks:

a) Identify all the referring expressions and their possible antecedents.

b) Apply the following constraints to resolve the references:

- Gender and number agreement
- Recency
- Grammatical role
- Semantic compatibility
- Discourse coherence

c) Construct a table showing:

Referring Expression	Possible Antecedents	Valid/Invalid	Reason
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d) Provide the final coreference chains.

e) Rewrite the paragraph by replacing pronouns with their correct references.

f) Discuss what ambiguity would arise if the sentence "He was looking for a book" appeared before introducing Arun.

2. Dialog Act Recognition and Response Generation

Conversation History:

Student: "I submitted my assignment, but I am worried that I may have made many mistakes."

Required Dialog Acts:

Reassure + Advise

Constraints:

1. Maintain entity coherence using assignment, mistakes, and you.
2. Use a Cause–Effect or Elaboration discourse relation.
3. Include at least two keywords from:
 - review
 - improve
 - confident
 - feedback
4. Response length must be 2–3 sentences.

5. Maintain a positive and polite tone.
6. Do not give contradictory advice.

Tasks:

- a) Identify the dialog acts required.
- b) Identify the important entities that must be maintained.
- c) Generate three possible responses satisfying all constraints.
- d) Compare the responses based on:
 - Coherence
 - Constraint satisfaction
 - Politeness
 - Relevance
- e) Select the best response and justify your answer.
- f) Explain what happens if entity coherence is not maintained.

3. Word Sense Disambiguation and Meaning Representation

Source Sentence:

"Priya sat on the bank and watched the boats moving across the water."

Note:

The word "bank" can refer to:

- A financial institution
- The side of a river

Constraints:

1. Resolve the correct word sense using contextual information.
2. Identify important entities and actions.
3. Represent the meaning using predicate logic.
4. Preserve semantic relationships.
5. Maintain coherence in the generated paraphrase.

Tasks:

- a) Identify the possible meanings of bank.
- b) Select the correct sense and justify your decision using contextual constraints.
- c) Write a predicate logic representation using suitable predicates such as:

sit(x)

bank(y)

beside(x,y)

watch(x,z)

boat(z)

move(z)

water(w)

d) Generate a simple English paraphrase without ambiguity.

e) Explain how Word Sense Disambiguation improves Machine Translation.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

DSA0306-NATURAL LANGUAGE PROCESSING

CO5 AT 1

B.VANI(192472348)

Q1. REFERENCE RESOLUTION AND DISCOURSE COHERENCE

a) Referring Expressions and Possible Antecedents

Paragraph:

"Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving."

Referring expressions:

1. "He" → Ravi or Arun
2. "him" → Ravi or Arun
3. "it" → the book
4. "they" → Ravi and Arun
5. "the book" → the previously mentioned book

b) Applying the Constraints

Gender and Number Agreement:

"He" and "him" refer to a singular male person. Both Ravi and Arun satisfy this constraint.

Recency:

"Arun" is the most recently mentioned male person before "He", so recency initially favors Arun.

Grammatical Role:

"Ravi" is the subject of the first sentence, while "Arun" is introduced as another participant. The subject continuity preference can favor Ravi as the topic of the next sentence.

Semantic Compatibility:

"looking for a book" is compatible with either person, so this constraint alone cannot completely resolve "He".

"He" is the person looking for the book, "him" is the person Arun helps, and "it" refers naturally to the book.

Discourse Coherence:

The paragraph is coherent if one person looks for the book and the other helps that person. A natural interpretation is Ravi is looking for the book and Arun helps Ravi.

c) Constraint Table

Referring Expression	Possible Antecedents	Valid/Invalid	Reason
He	Ravi	Valid	Male, singular, grammatical subject continuity
He	Arun	Valid	Male, singular, recent mention
him	Ravi	Valid	Male, singular and fits the helping relation
him	Arun	Invalid	Arun cannot naturally help himself in this interpretation
it	book	Valid	Singular object and semantic compatibility
it	Ravi	Invalid	Person cannot be the object being found
it	Arun	Invalid	Person cannot be the object being found
they	Ravi + Arun	Valid	Plural reference to both participants
the book	previously mentioned book	Valid	Maintains discourse continuity

d) Final Coreference Chains

A coherent resolution is:

Ravi → He → him

Arun → Arun

book → it → the book

Ravi + Arun → they

Therefore:

He = Ravi

him = Ravi

it = book

they = Ravi + Arun

the book = same book

e) Rewritten Paragraph

"Ravi met Arun at the library. Ravi was looking for a book on Artificial Intelligence. Arun helped Ravi find the book. Later, Ravi and Arun discussed the book before leaving."

f) Ambiguity if "He was looking for a book" Appears Before Introducing Arun

If the sentence "He was looking for a book" appears before Arun is introduced, there is no previously introduced male antecedent for "He". The pronoun would have no clear discourse reference.

For example:

"He was looking for a book. Ravi met Arun at the library."

Here, the identity of "He" cannot be reliably determined from the preceding discourse. This creates a coreference ambiguity and reduces text coherence.

Result

Constraint-based reference resolution uses agreement, recency, grammatical role, semantic compatibility, and discourse coherence to identify the most suitable antecedents. In this paragraph, Ravi is the most coherent antecedent for "He" and "him", while "it" and "the book" refer to the book.

Q2. DIALOG ACT RECOGNITION AND RESPONSE GENERATION

a) Required Dialog Acts

The required dialog acts are:

1. Reassure
2. Advise

The response should first reduce the student's worry and then provide useful advice.

b) Important Entities

The important entities are:

assignment

mistakes

you

These entities must remain consistent throughout the response.

c) Three Possible Responses

Response 1:

"You may have made some mistakes in the assignment, but that does not mean it is a failure. Review your assignment carefully, use the feedback to improve, and you can feel more confident."

Response 2:

"It is normal to worry after submitting an assignment, and mistakes can be useful for learning. Review the assignment when possible, use the feedback to improve, and become more confident next time."

Response 3:

"Do not worry too much about the assignment, because making mistakes is part of learning. Review your work and use feedback to improve so that you feel confident about your next submission."

d) Comparison

Response	Coherence	Constraint Satisfaction	Politeness	Relevance
Response 1	High	High	High	High
Response 2	Very High	High	High	Very High
Response 3	High	High	High	High

e) Best Response

"It is normal to worry after submitting an assignment, and mistakes can be useful for learning. Review the assignment when possible, use the feedback to improve, and become more confident next time."

Justification:

1. Contains both required dialog acts: reassurance and advice.
2. Maintains the entities assignment, mistakes, and you.
3. Contains the required keywords "review", "feedback", "improve", and "confident".
4. Uses a positive and polite tone.
5. Uses a cause-effect relationship between mistakes and learning.
6. Contains exactly two sentences.

f) Effect of Breaking Entity Coherence

If entity coherence is not maintained, the response may refer to the wrong person or object. The system may suddenly discuss another assignment, a different person, or

unrelated mistakes. This makes the response confusing and reduces the reliability of the conversational agent.

Result

The best response satisfies the dialog-act, entity, discourse, keyword, length, and politeness constraints while maintaining a clear connection to the student's concern.

Q3. WORD SENSE DISAMBIGUATION AND MEANING REPRESENTATION

a) Possible Meanings of "Bank"

The word "bank" has two possible meanings:

1. Financial institution
2. Side of a river

b) Correct Sense and Justification

Sentence:

"Priya sat on the bank and watched the boats moving across the water."

The correct sense is:

bank = side of a river

Reason:

1. "boats" is strongly associated with a river or water body.
2. "water" provides a strong contextual cue.
3. "sat on the bank" naturally describes sitting beside a river.
4. The financial meaning of bank is not compatible with watching boats moving across water.

Therefore, contextual constraints clearly favor the river-bank sense.

c) Predicate Logic Representation

Let:

Priya = p

Bank = b

Boat = z

Water = w

Predicate representation:

sit(Priya,b)

bank(b)

beside(Priya,b)

watch(Priya,z)

boat(z)

move(z)

water(w)

across(z,w)

A combined representation is:

$\text{bank}(b) \wedge \text{sit}(\text{Priya}, b) \wedge \text{beside}(\text{Priya}, b) \wedge \text{boat}(z) \wedge \text{watch}(\text{Priya}, z) \wedge \text{move}(z) \wedge \text{water}(w) \wedge \text{across}(z, w)$

This represents that Priya sat beside the river bank and watched a boat moving across the water.

d) Simple English Paraphrase

"Priya sat beside the river and watched the boats moving across the water."

This paraphrase removes the ambiguity of "bank" by explicitly using "river".

e) How WSD Improves Machine Translation

Word Sense Disambiguation determines the intended meaning of an ambiguous word before translation. This prevents the translation system from choosing an incorrect sense.

For example:

bank = river bank

must be translated according to the geographical meaning rather than the financial meaning.

WSD uses contextual words such as "boats" and "water" to select the correct sense. This produces more accurate and coherent translations and paraphrases.

Result

WSD correctly selects the river-bank meaning using contextual constraints. Predicate logic preserves the relationships among Priya, the bank, boats, and water, while the paraphrase makes the intended meaning explicit.